Families of Elliptic Curves over $k(t)$, $\mathbb{F}_q(t)$, $\mathbb{C}(t)$: Height Moduli, Exact Counts, Mordell–Weil Rank Jumps

Abstract. The notion of height does more than order the infinite set of elliptic curves: it turns their totality into a geometric space. After a brief motivation over $\mathbb{Q}$, I will explain how this geometry connects moduli, exact counting, and Mordell–Weil rank jumps.

Over $k(t)$, an elliptic curve can be viewed as an elliptic surface over $\mathbb{P}^1_k$. Its Faltings height is the degree of the fundamental line bundle; on the semistable locus, this is the degree of the Hodge line bundle pulled back along the classifying map to $\overline{\mathcal{M}}_{1,1}$. The resulting height moduli stack is of finite type and parametrizes elliptic curves of fixed Faltings height n, with a natural correspondence between its strata and Kodaira fibre configurations.

Over $\mathbb{F}_q(t)$, motivic methods, together with explicit calculations in characteristics 2 and 3, yield exact weighted and unweighted counts of elliptic curves of bounded height in every characteristic. The lower-order terms reflect nonsmooth-locus corrections, extra-automorphism loci, and minimality defects.

Over $\mathbb{C}(t)$, when restricted to elliptic surfaces with only irreducible nodal singular fibres, the Shioda–Tate formula identifies Mordell–Weil rank jumps with Picard rank jumps. At each fixed height $n \geq 2$, the intersection of the period image with Hodge loci produces analytically dense loci of Jacobian elliptic surfaces with Mordell–Weil rank at least r throughout the transverse range

$$1 \leq r \leq \left\lfloor \frac{10n-2}{n-1} \right\rfloor$$

A second-order period calculation gives a direct proof of exact-rank-10 density for $n \geq 3$.